

fn sample_geometric_exp_slow

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Proves soundness of `fn sample_geometric_exp_slow` in `mod.rs` at commit `1f9230c` (outdated¹). This proof is adapted from subsection 5.2 of [CKS20].

1 Hoare Triple

Precondition

Compiler-verified

Argument `x` is of type `RBig`, a bignum rational

Caller-verified

`x > 0`

Pseudocode

```
1 def sample_geometric_exp_slow(x) -> int:
2     k = 0
3     while True:
4         if sample_bernoulli_exp(x): #
5             k += 1
6         else:
7             return k
```

Postcondition

Theorem 1.1. For any setting of the input parameter `x` such that the given preconditions hold, `sample_geometric_exp_slow` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed as `Geometric(1 - exp(-x))`.

Definition 1.2. If $K \sim \text{Geometric}(p)$, then for $k \in \{0, 1, \dots\}$

$$\Pr[K = k] = (1 - p)^k \cdot p. \quad (1)$$

Definition 1.3. If $B \sim \text{Bernoulli}(p)$, then for $b \in \{\top, \perp\}$

$$\Pr[B = b] = \begin{cases} p & b = \top \\ 1 - p & b = \perp \end{cases} \quad (2)$$

Assume the preconditions are met.

¹See new changes with `git diff 1f9230c..e376a25 rust/src/traits/samplers/cks20/mod.rs`

Lemma 1.4. `sample_geometric_exp_slow` only returns `Err(e)` when there is a lack of system entropy.

Proof. The preconditions on `x` satisfy the preconditions on `sample_bernoulli_exp`, so by its definition, it only returns an error if there is a lack of system entropy. The only source of errors is from this function, therefore `sample_geometric_exp_slow` only returns `Err(e)` when there is a lack of system entropy. $\square$

Theorem 1.5. [CKS20] If the outcome of `sample_geometric_exp_slow` is `Ok(out)`, then `out` is distributed as $\text{Geometric}(1 - \exp(-x))$.

Proof. The distribution of the i^{th} boolean returned on line 4 is $B_i \sim \text{Bernoulli}(\exp(-x))$, because the preconditions on `x` satisfy the preconditions for `sample_bernoulli_exp`.

$$\begin{aligned} \Pr[\text{out} = k] &= \Pr[B_1 = B_2 = \dots = B_k = \top \wedge B_{k+1} = \perp] \\ &= \Pr[B_{k+1} = \perp] \prod_{i=1}^k \Pr[B_i = \top] && \text{All } B_i \text{ are independent.} \\ &= (1 - \exp(-x)) \exp(-x)^k \end{aligned}$$

By Definition 1.2, setting $p = 1 - \exp(-x)$, then `out` $\sim \text{Geometric}(1 - \exp(-x))$. $\square$

Proof of Theorem 1.1. Holds by 1.4 and 1.5. $\square$

References

- [CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.